

GATE

Computer Science & IT



Compiler Design



Intermediate Code Generation

Lecture No. 01



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Recap of Previous Lecture



Topic

Parsers

Topic

Topic

Topics to be Covered



Topic

Topic

Topic

$\text{closur}(I)$

Closure(I) where I is a set of LR(1) items

I is a set of LR(1) items like $[A \rightarrow \alpha \cdot B\beta, \$]$, $[C \rightarrow a \cdot, a]$

The function closure as i/p, a set of LR(1) items

and produces Set of LR(1) items as o/p. The function

is evaluated using following rules

1) Initially add every item from i/p to o/p

2) If $A \rightarrow \alpha \cdot B\beta, \$$ is in I where B is a non terminal
and $B \rightarrow \gamma$ is the rule for B then add $B \rightarrow \gamma, \text{First}(\underline{B\beta})$ ✓
to closure(I). Repeat this for every newly added item.

Goto(I, x):

Given the set of all items of the form

$[A \rightarrow \alpha \cdot \underline{x} \beta, \underline{a}]$ in I ,

$$\text{goto}(I, x) = \underline{\text{closure}([A \rightarrow \alpha \underline{x} \cdot \beta, \underline{a}])}$$

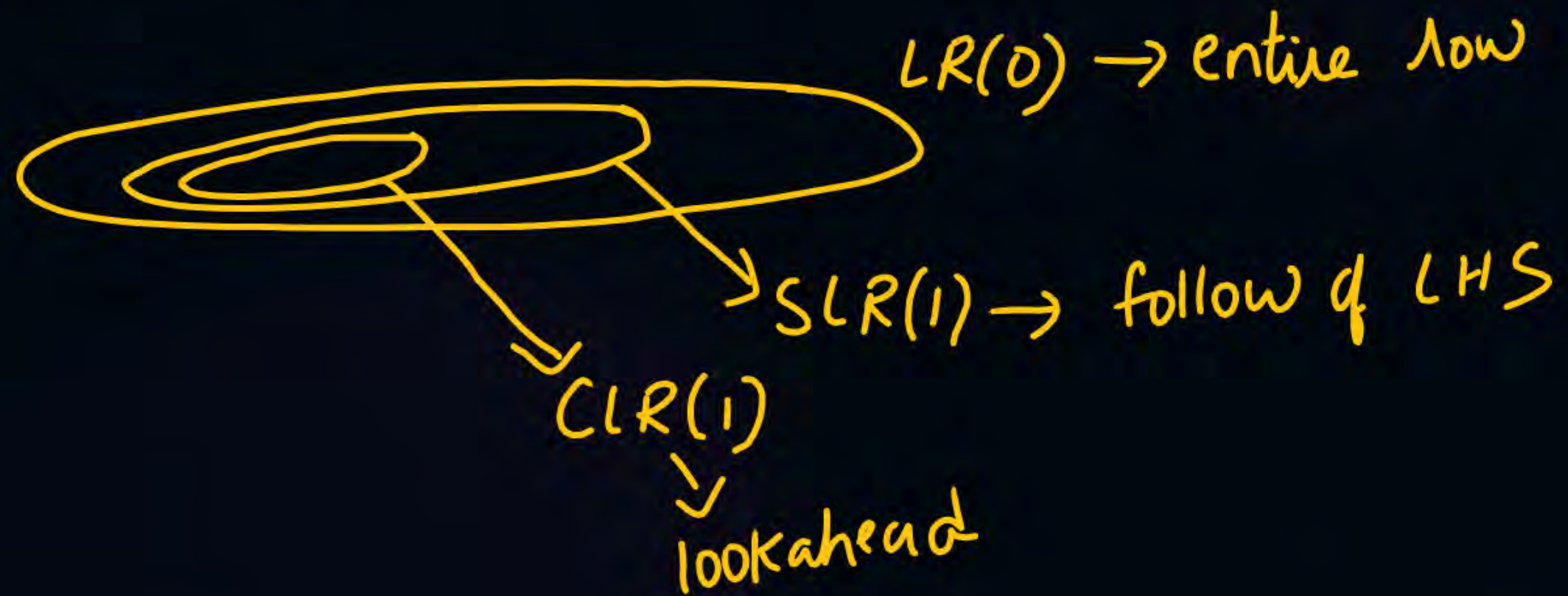
The goto function involves two steps

1) Find transition ✓✓

2) Apply closure ✓✓

→ while finding closure, look ahead may change

Advantage of CLR(1)

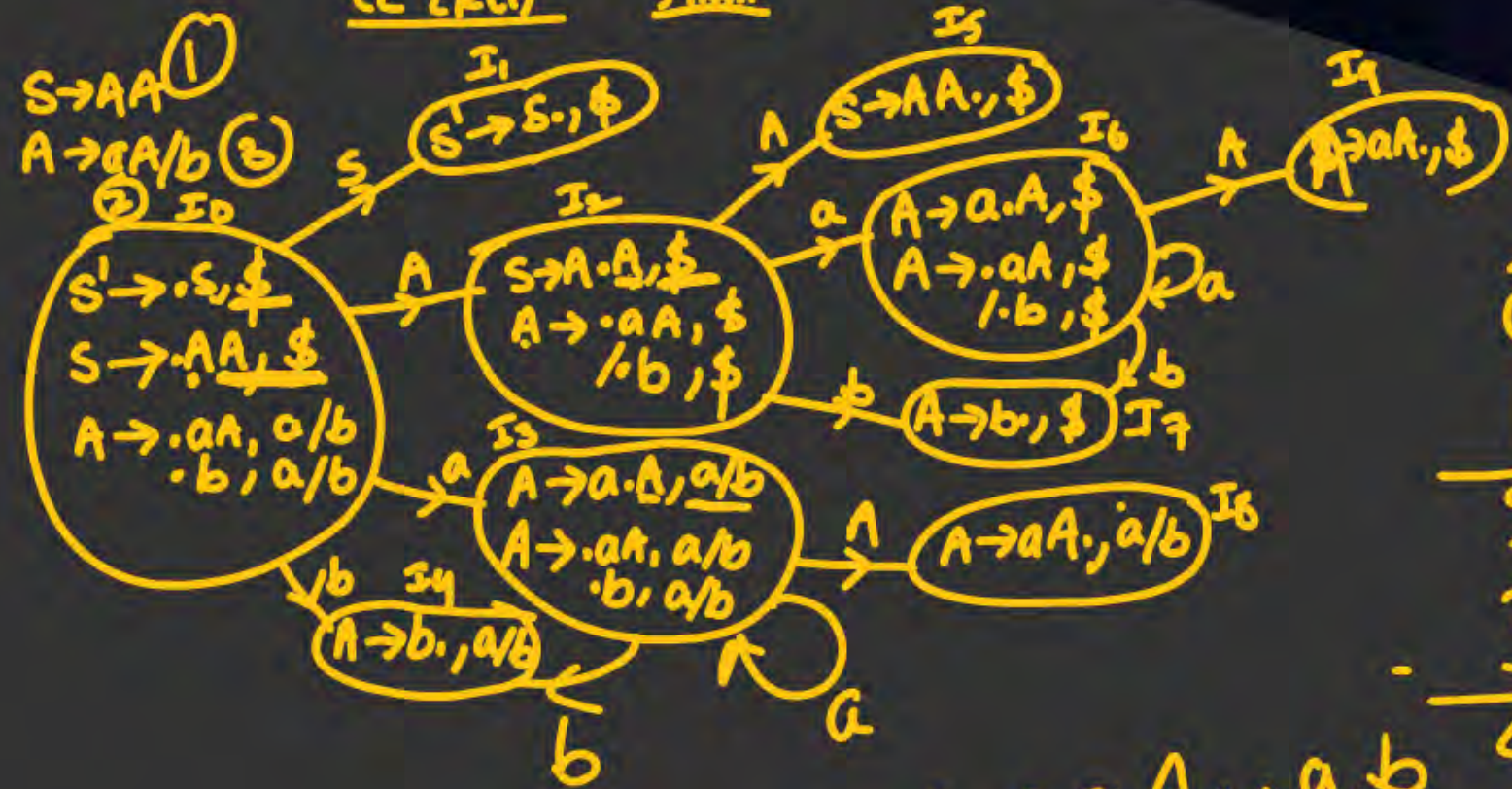


$CLR(1) \Delta LR(1)$

Blank spaces in $CLR(1)$ increases which will increase the end detecting capacity.

So $CLR(1)$ is more powerful
The num of states in $CLR(1)$ may be more than other LR parres,
which will increase blank spaces and help in end detecting

CC LR(1) 5 min



CLR(1) parsing table

	action.			Goto	
	a	b	\$	S	A
0	S_3	S_4		1	2
1			acc		
2	S_6	S_7			5
3	S_3	S_4			8
4	γ_3	γ_3			
5			γ_1		
6	S_6	S_7			9
7			γ_3		
8	γ_2	γ_2			
9			γ_2		

$A \rightarrow aA \cdot, a, b$
 γ_2

$A \rightarrow b \cdot, a/b$

γ_3

$S \rightarrow AA \cdot, \$$

γ_1

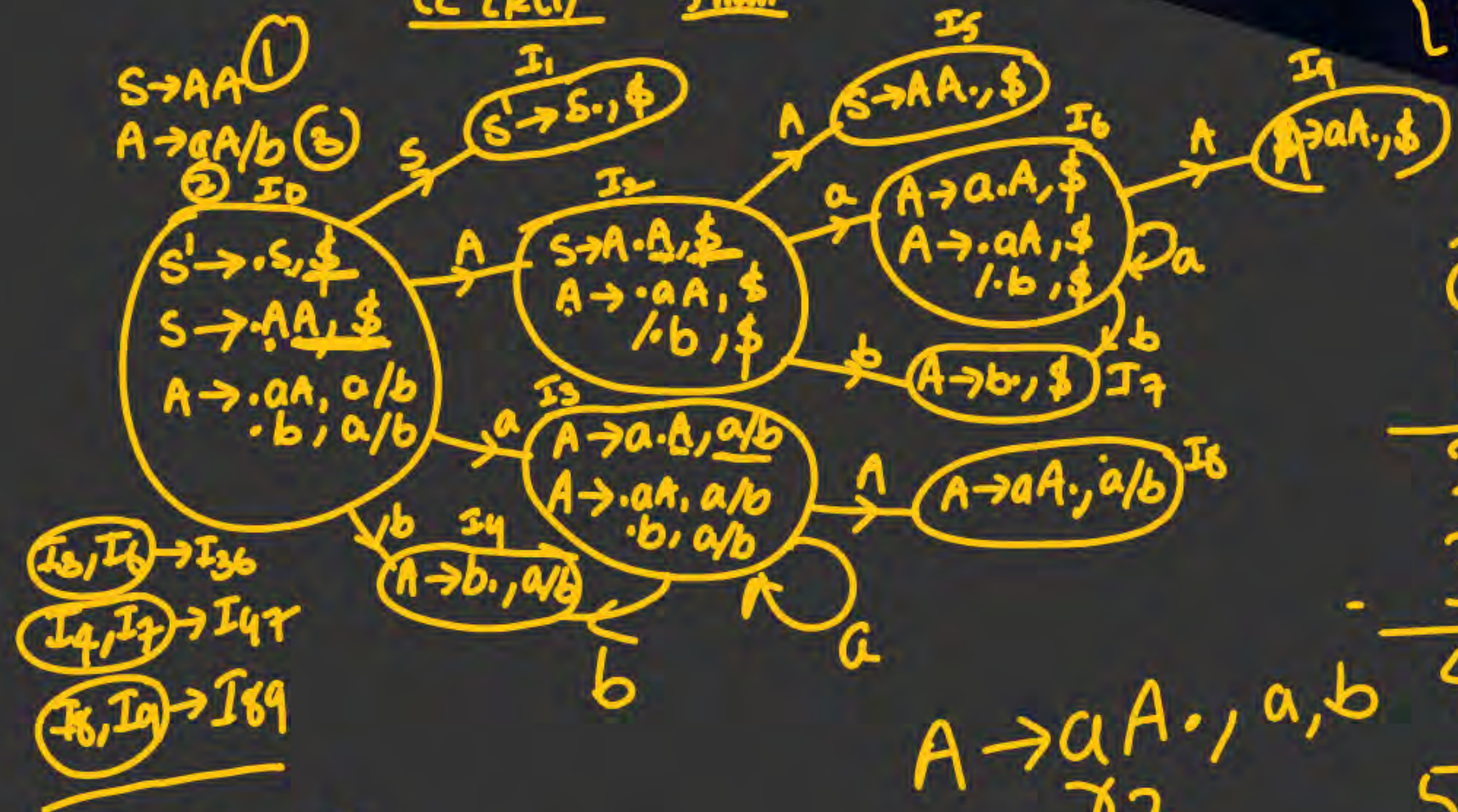
$A \rightarrow b \cdot, \$$

γ_3

$A \rightarrow aA \cdot, \$$

γ_2

CC LR(1) 5 min



~~LAUR(1) LR(1)~~ parsing table

	action.			Goto	
	a	b	\$	S	A
0	S ₃₆	S ₄₇		1	2
1			acc		
2	S ₃₆	S ₄₇			5
36	S ₃₆	S ₄₇			89
47	r ₃	r ₃	r ₃		
5			r ₁		
36	S ₃₆	S ₄₇			89
47			r ₃		
89	r ₂	r ₂	r ₂		
89			r ₂		

$A \rightarrow aA \cdot, a, b$
r₂

$A \rightarrow b \cdot, a/b$
r₃

$A \rightarrow b \cdot, \$$
r₃

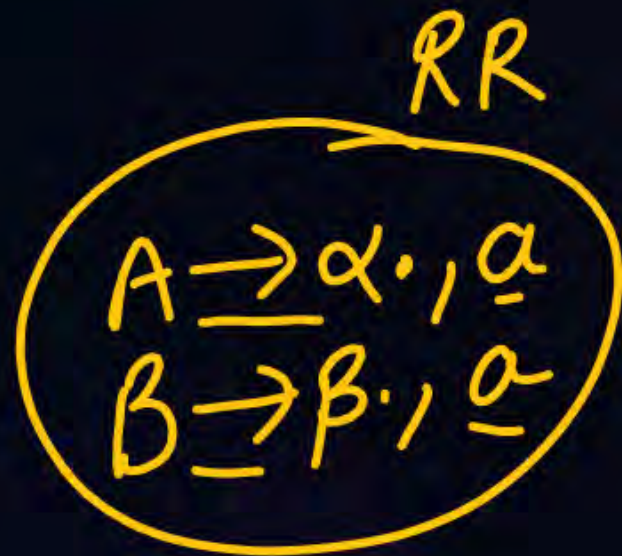
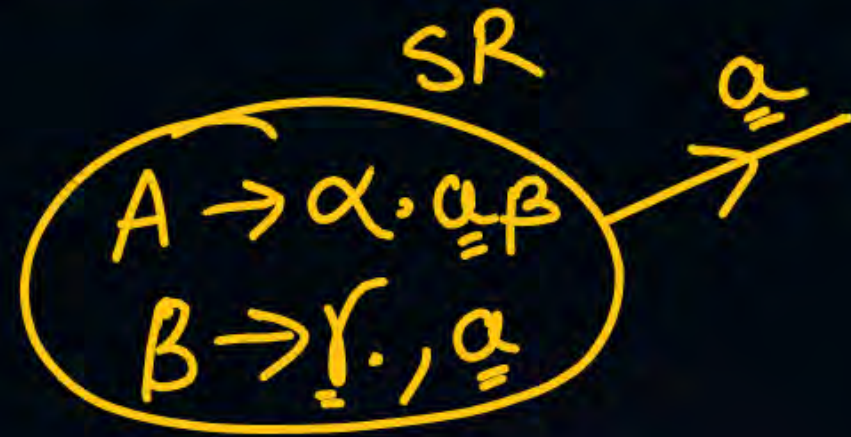
$S \rightarrow AA \cdot, \$$
r₁

$A \rightarrow aA \cdot, \$$
r₂

CLR(1)=10
LAUR(1)=7
SLR(1)=7
LR(0)=7

CLR(0) ≥ SLR(1) = LAUR(1) = LR(0)

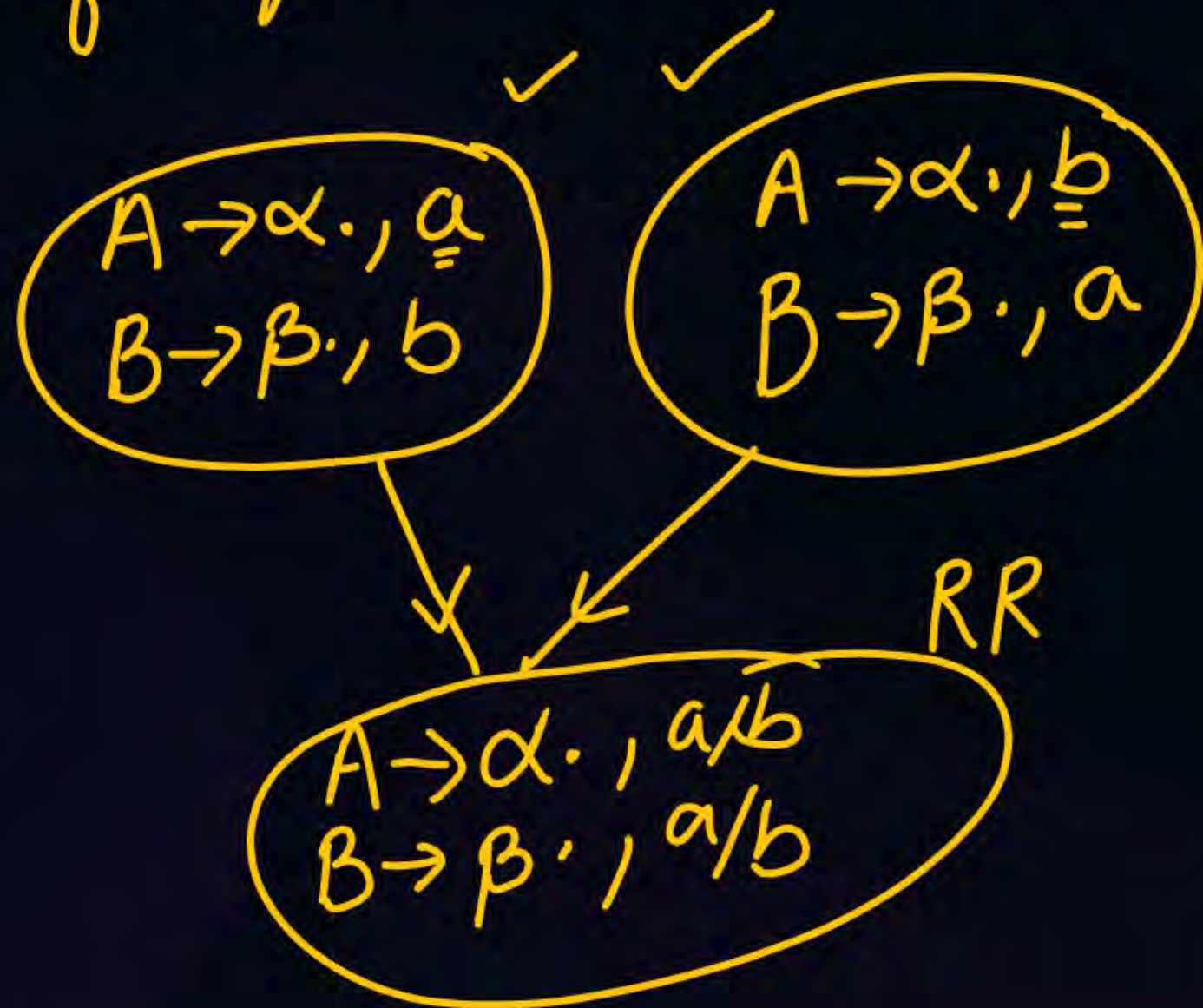
Conflicts in LR(0) items



If above conflicts are present then grammar is not CLR(1) 

If grammar is not CLR(1), then it is not LALR(1) also.

Even if a grammar is CLR(1), it may not be LALR(1)





→ If there are no SR Conflicts in $CLR(1)$, then there will be no SR Conflicts in $LALR(1)$.

→ If there are no RR Conflicts in $CLR(1)$, then there may be RR Conflicts in $LALR(1)$

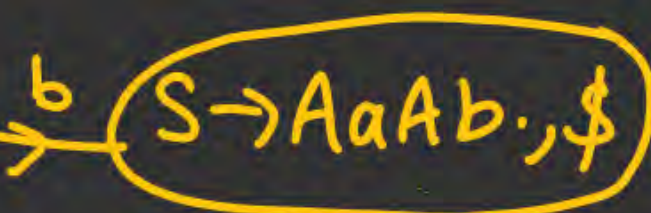
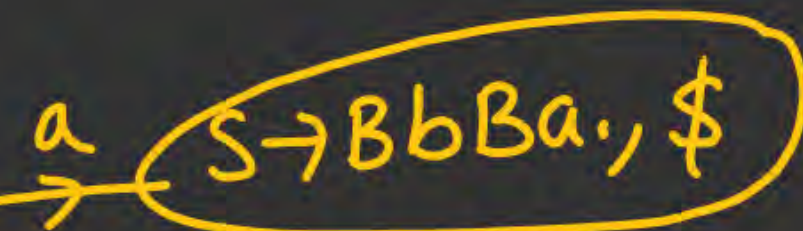
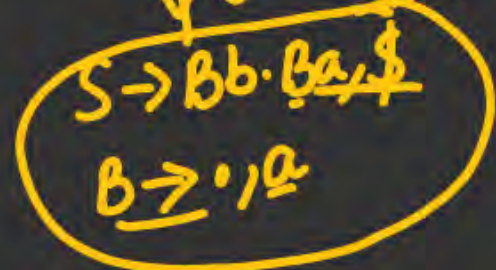
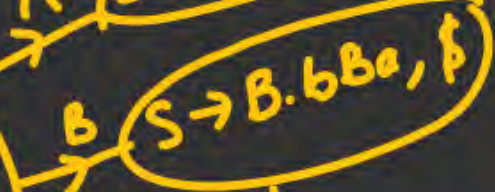
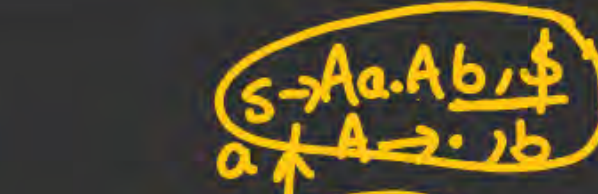
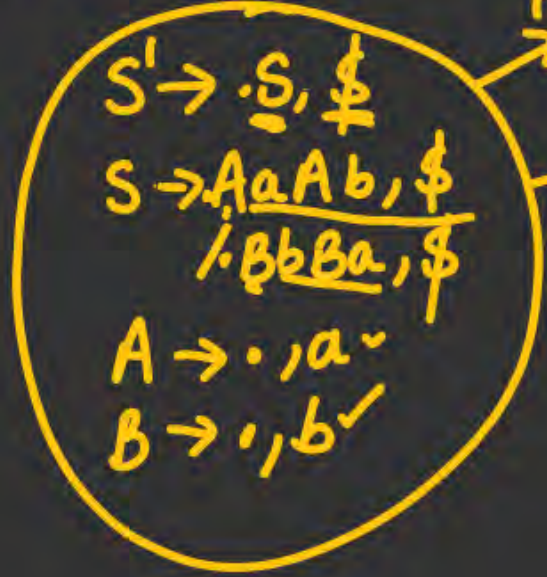
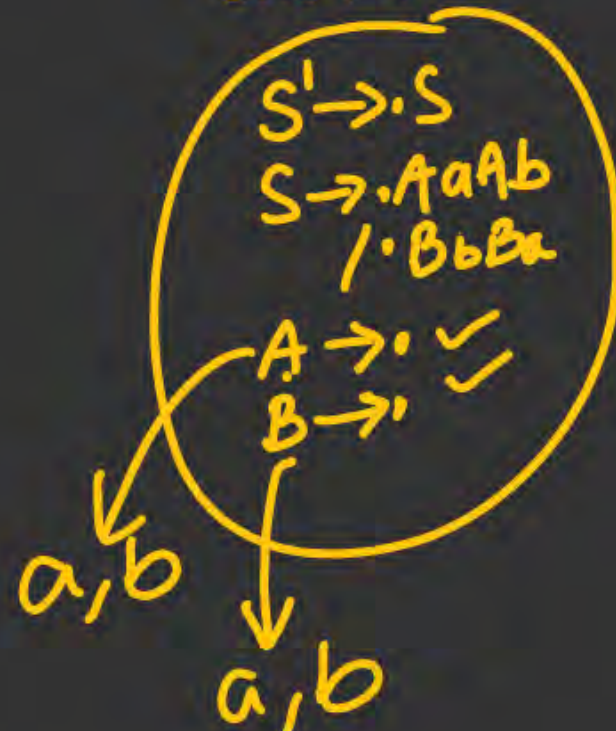
$S \rightarrow AaAb.$
 $/ BbBa.$
 $A \rightarrow \epsilon$
 $B \rightarrow \epsilon$

$LL(1) \checkmark$
 $LR(0) \times$
 $SLR(1) \times$
 $CLR(1) \checkmark$
 $LALR(1) \checkmark$

CLR(1)

LR(1) item

LR(0) item



$CLR(1) = LALR(1)$

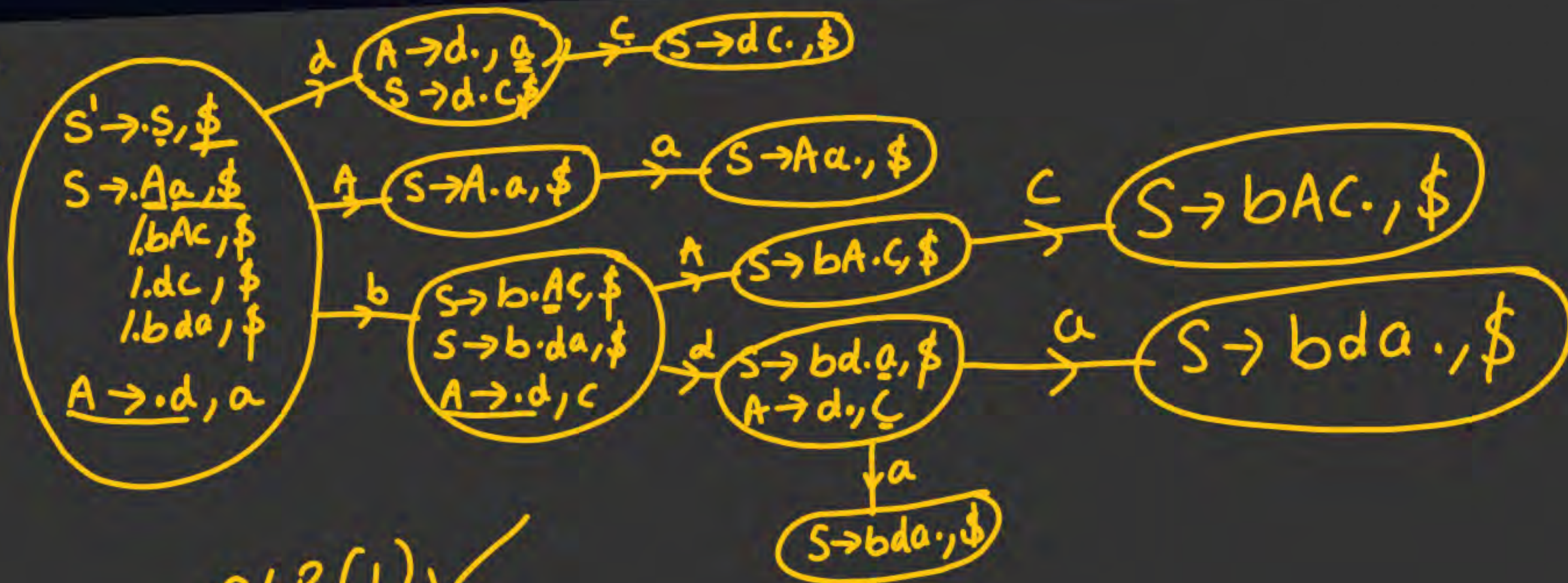
$S \rightarrow Aa - d$
 $/bAc$
 $/dc - d$
 $/bda$

$A \rightarrow d$

$LL(1) \times$
 $LR(0) \times$
 $SLR(1) \times$
 $CLR(1)$
 $LALR(1)$

5 minutes

$S \rightarrow Aa - d$
 $/bAc$
 $/dc - d$
 $/bda$
 $A \rightarrow d$



CLR(1) ✓

LALR(1)

CLR(1) = LALR(1)

$S \rightarrow Aa$
 $/bAc$
 $/Bc$
 $/bBa$

$A \rightarrow d$
 $B \rightarrow d$

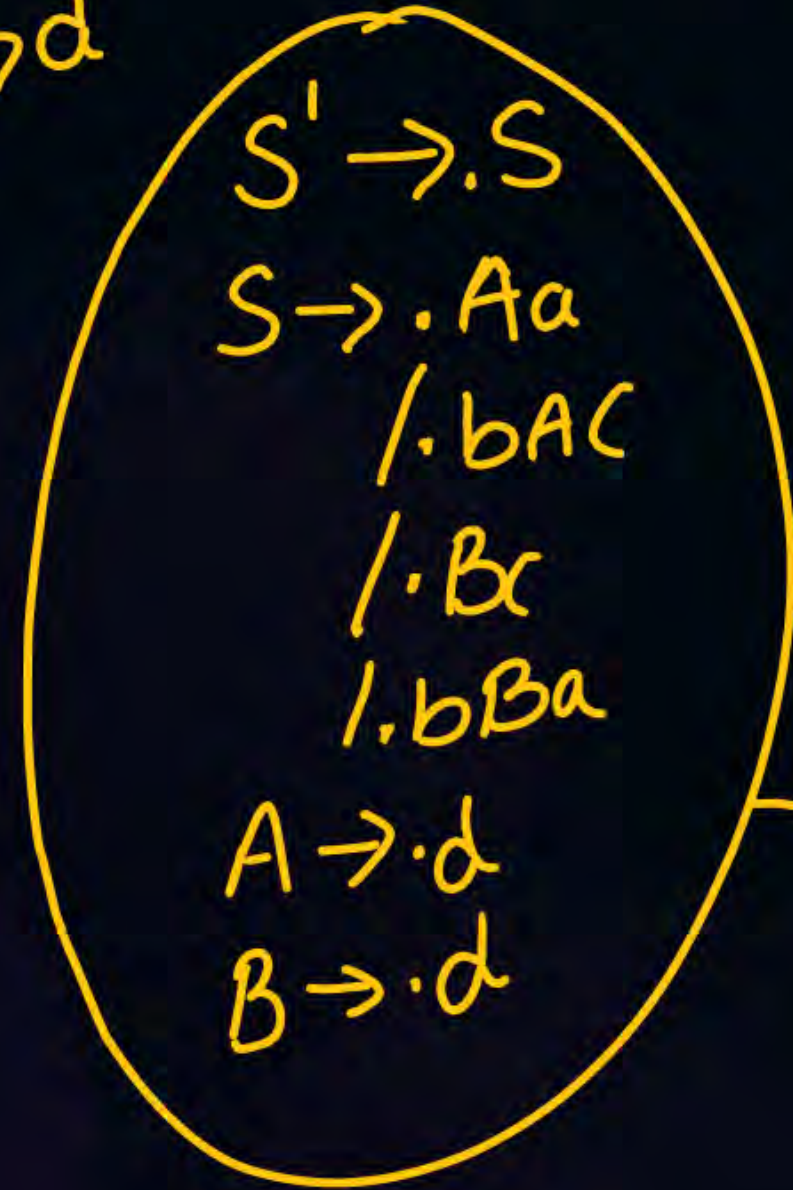
$LL(1)$
 $LR(0)$
 $SLR(1)$
 $CLR(1)$
 $LALR(1)$

5 min

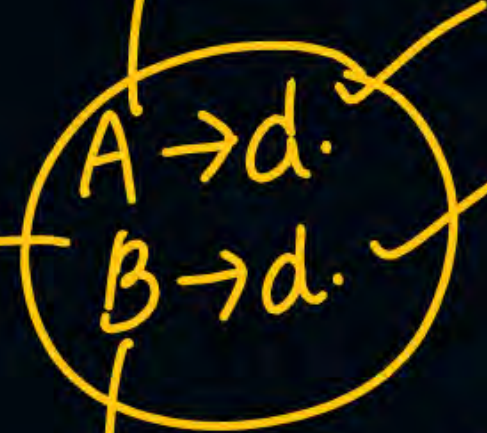
$S \rightarrow Aa \rightarrow d$ $LL(1) \times$
 $LR(0)$

$/bAC$
 $/BC \rightarrow d$
 $/bBa$

$A \rightarrow d$
 $B \rightarrow d$



d



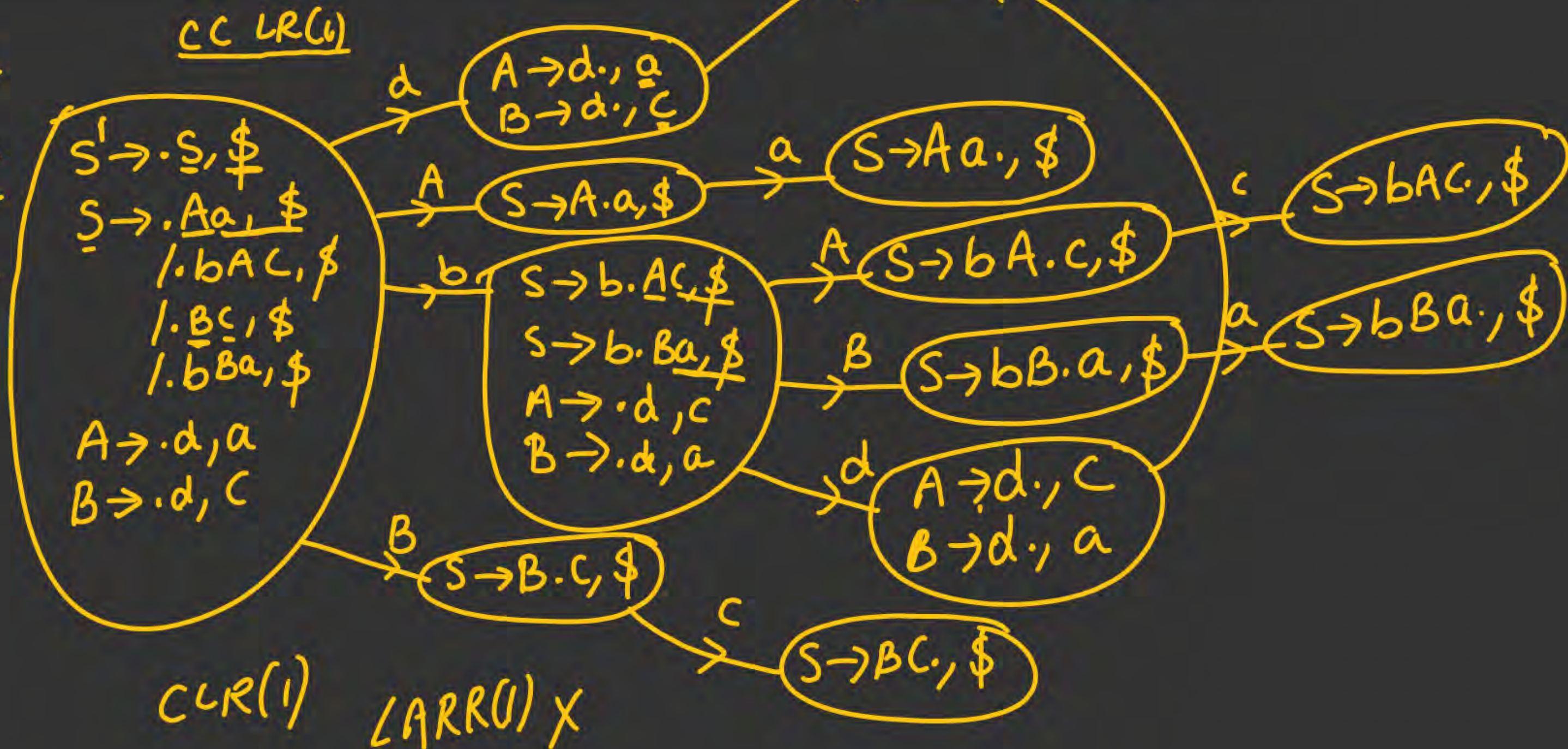
$Foll(A) = \{a, c\}$

$Foll(B) = (\underline{a}, \underline{c})$

$LR(0) \times$

$SLR(1) \times$

$S \rightarrow Aa$
 $S \rightarrow bAC$
 $S \rightarrow BC$
 $S \rightarrow bBa$
 $A \rightarrow d$
 $B \rightarrow d$



✓
 $S \rightarrow A$
 $A \rightarrow \underline{A}B/\epsilon$
 $B \rightarrow aB/b$

favdite

HW

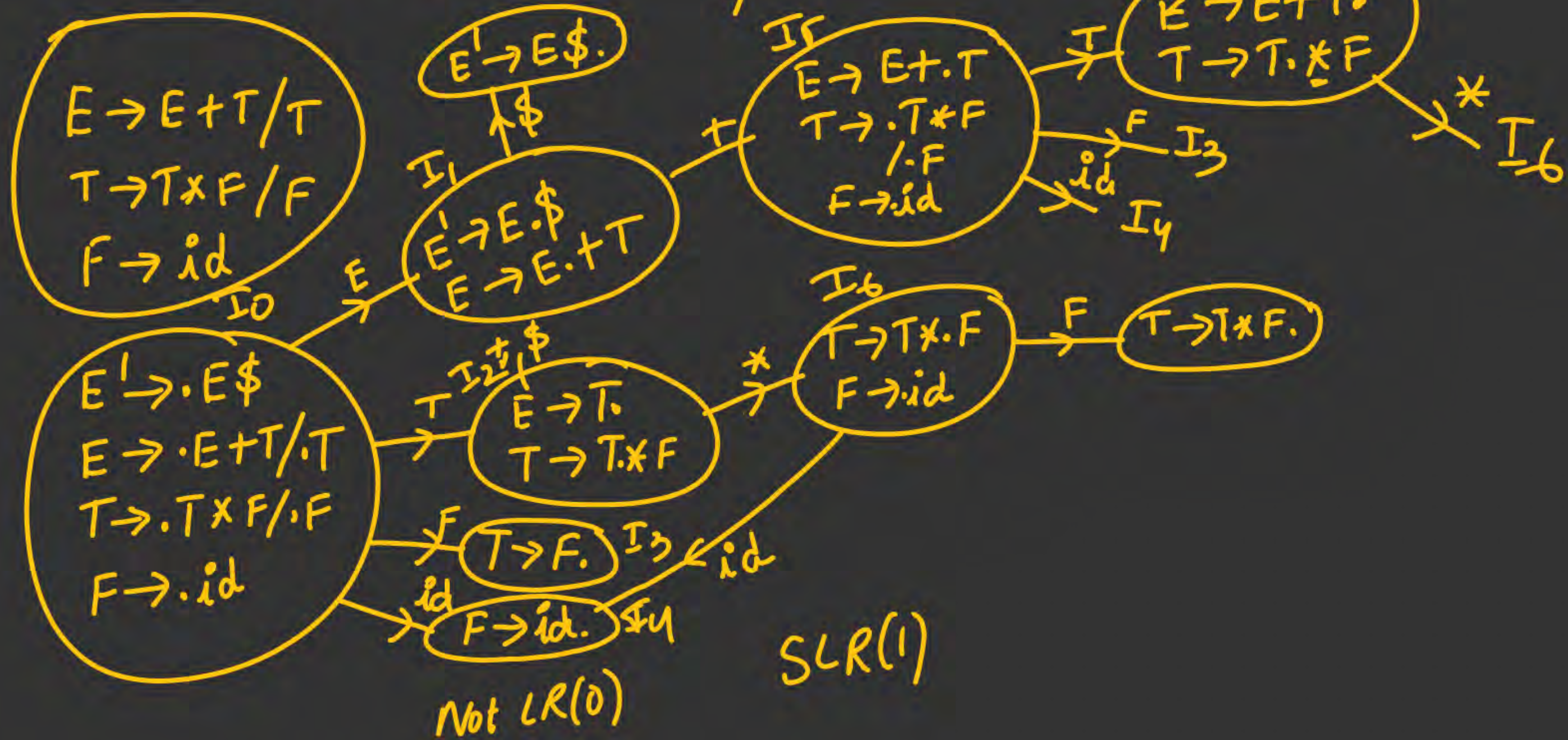
difficult

< 100

$E \rightarrow E + T / T$	LL(1) X
$T \rightarrow T * F / F$	LR(0)
$F \rightarrow id$	SLR(1)
	CLR(1)
	LALR(1)

5 min

CLR(1) LALR(1)



→ Consider $\overset{LR(0)}{SLR(1)}$ and $\overset{LR(0)}{LALR(1)}$ table for a CFG

X a) Goto both tables may be different

b) Shift entries are identical in both tables

c) Reduce entries in table may different

d) end entries in tables may be different

which are false

$$\begin{array}{ccc}
 n(SLR(1)) & = & n(LALR(1)) \\
 \downarrow & & \downarrow \\
 CC\ LR(0) & & CC\ LR(1) \\
 & & \downarrow \\
 & & LR(0)
 \end{array}$$



THANK - YOU